

Review Study on Software Defect Prediction Models premised upon Various Data Mining Approaches

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Abstract - Software Reliability is becoming a crucial aspect for current software systems. Reliability is very essential factor of software quality as this factor computes the failures of the software. Software defect prediction models are gaining important role in attaining high reliable software system. These models contribute in finding faults early and also help in removing these faults to produce efficient and reliable software. For many years, Factual studies and approaches have been contemplated. This paper reviews some defect prediction models premised upon various data mining approaches. This paper also overviews neural networks concept which is considered as one of the rising approaches for defect prediction models.

Keywords----Approaches of Data mining; Software defect & defect prediction models; Role of Neural Networks

I. INTRODUCTION

Despite of thorough planning, proper documentation as well as appropriate controlling of the process during the software development phase, the existence of various defects are unpredictable[1]. These faults are basically defined as the variation from expected software operation that gives rise to software failures and even degrades the software quality. In the present vanguard competition, it is very important to generate vigilant attempts to regulate as well as remove these software defects [1]. Detection of defects at an early stage of software evolution will help to improve adequate cost as well as increase customer satisfaction. Detecting faults at early phase will also increase software reliability. Software defect prediction focus on developing models for various software projects. These models prove helpful in reducing efforts required during locating software defects[2]. These models incorporates independent variables i.e. software metrics which are collected and measured during software development phase as well as includes dependent variables i.e. Faulty or Non-Faulty[3]. In current defect prediction models, complexity as well as size based metrics are used for anticipating those software faults that can occur during operation or testing phase of software project[1]. Whereas other defect prediction models, especially reliability based models needs working profile of a software system to anticipate rate of defects that can occur during any software project[1].

A. General SoftwareDefect Prediction Process [1]:

For constructing a software defect prediction model [1], there is a requirement of fault and measurement related data extracted from definite software progress efforts that can

beused as training set. A common and general software defect prediction process shown below has been described by Sunghun Kim et al. [4].

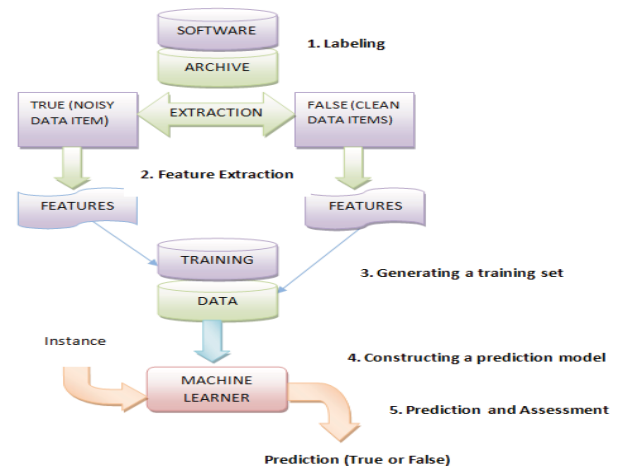


Fig. 1. Software Defect Prediction Process [1]

(1)Labelling[4]: Defect data which is gathered during this phase act as a training data for defect prediction model[4].This phase includes gathering data elements from software repository and then assigning them with the values-TRUE or FALSE.

(2)Excerptingfeatures and building training corpus [4]: This phase excerpts features for anticipating labels of data elements. By linking data items and features of data elements, a training corpus.

(3)Constructing prediction models [4]: Various machine learners like Support Vector Machines (SVM) and Bayes Net with considering training corpus as base can be used to frame a defect prediction model.

(4)Evaluation [4]: There is a requirement of a testing data set for evaluation of a software defect anticipation model. We anticipate the labels of data items which are the part of the experimentation set and even access defect anticipation model by correlating anticipated labels and actual labels.

II. DATA MINING AND ITS TECHNIQUES

Data Mining means extorting meaningful input or knowledge from a distributed data and even employs various

analytical tools [14] to extort patterns and information from large distributed data.

Data Mining can be classified into-Predictive as well as Descriptive tasks. The conclusive target of data mining techniques lies in prediction and predictive data mining technique is the most familiar data mining technique [15]. This technique is based on formulas that compares success and failures and these formulas are used to predict future results.

Predictive Modelling is technique by which model is created to anticipate the result. If the result is absolute it is known as classification and if it is analytical it is known as regression[16].Detailed modelling or clustering is the process of assigning observations into clusters so that the observations of same cluster are also similar[16].

Data mining approaches that can be a part of prediction process are as follows:

A. Regression[17]:

It is a statistical technique to analyse inter-relationship between dependent [17] and independent [17] variables. This interrelationship is asserted with the help of equation that evaluates the dependent variable [17] as linear behaviour of predictor variable [17].

Linear Regression Equation [17]:

$$Y=a+bX+u \quad [17] \quad (1)$$

B. Association Rule Mining [17]

This technique is used for determining interesting relationships within variables of massive databases. This technique basically focuses on discovering rules that will evaluate the item based occurrence on the other items occurrence.

C. Clustering[17]

This process deals with categorizing group of items into clusters consisting of similar kind of members. It is a technique by which the group of items are categorize in such a fashion that elements of same cluster are identical to each other but different to further clusters.

D. Classification[17]

It is a process of evaluating a certain outcome on the basis of input data. This technique takes data also was known as learning data in which all items are identified by known class identifiers [17]. The main motive of this technique is analyzing and learning from learning data and then constructing model on basis of this learning data.

1) Neural Networks[17]

They are a non-linear model which learns from training data and are similar to biological neural networks. They consist of interconnected nodes known as neurons that work within the network in parallel to produce results.

2) Decision Trees[17]

This predictive technique produces both classification [17] as well as regression [17] models which have tree-like

architecture. It also indicates stratified structure representing decisions and their outcomes [17].

3) Naïve Bayes[17]

They are premised upon Bayes theorem which considers presumption within predictors. It is based upon the presumption that the existence or non-existence of peculiar trait of a class is not associated with the existence or non-existence of any another trait [17].

a) SVM(Support Vector Machines)[17]

They are premised upon conception of decision planes [17] that determine decision boundaries. A decision plane separates various set of objects possessing different class membership.

b) Case Based Reasoning[17]

It means solving out new problems based upon similar past problems [17].In these technique new unclassified records are compared with familiar patterns.

III. LITERATURE REVIEW

There has been a tremendous progress of continuous research in field of assessment of effective software reliability due to increased complexities of software as well as demand of reliable software. Prediction of software defects is not a new term in software engineering discipline. In recent years many researchers have recommended various Software Defect Prediction models as well as techniques.

A. Defect Prediction models using Classification techniques

Author in [2] presented various clustering techniques that help developers to identify defects considering existing software metrics .These techniques proved a better solution to improve software quality.

Authors in [5] have discussed various methodologies for Software Defect Prediction [5].This has also depicted various victimization classification techniques.

Authors in [6] proposed the usage of neural network for anticipating software quality of large telecommunication systems and classified modules [6] into fault prone or non-fault prone [6].

In [7],analysts uses metrics based on software size part of requirement analysis phase for anticipating software defects using fuzzy logic technique as well as investigated the anticipative quality of the suggested approaches based on qualitative data collected by requirement metrics[7] of 20 software projects[7].

B. Defect Prediction models using Clustering techniques

In [8], authors suggested a software defect prediction technique established on functional clusters for elaborating the performance of software defect prediction [8].They found that precision (7.8% to 91.6%) of software defect prediction was improved using cluster based prediction models.

In the paper [9], K-means clustering technique has been used for discovering fault proclivity of Object-Oriented based systems [9].In this paper, Metric values of JEdit software has

been used to generate rules for categorizing software modules into Faulty as well as Non-Faulty modules [9].

C. Defect Prediction[17] models premised upon Association Rule Mining[17] Technique

In the paper [10], an empirical assessment for the suggested model on wide source NASA datasets [10] has been performed.

In [11], authors have used association rule mining[11] approaches for anticipating defect associations[11] as well as defect correction[11]. The methods like PART[11], C4.5[11] and Naive Bayes[11] were used to compare software defect correction effort prediction[11] methods and analysed that accuracy was improved by at least 23%.

D. Defect Prediction models using Hybrid Technique

In the paper [12], Genetic Algorithm (GA) has been presented and experimental study has been carried out to compare GA-DT based hybrid technique with DT classification based technique.

In the paper [13], a fault-prone[13] module prediction[13] technique, combination of association rule mining[13] as well as logistic regression analysis[13] has been suggested by researchers. The prediction performance [13] of the suggested model was [13] analysed and correlated with 3 other fault-prone [13] models-logistic regression model [13], linear discriminant model [13] and classification tree [13].

IV. DISCUSSION[17]

On analyzing literature, different machine learning techniques like supervised [17] as well as unsupervised learning [17] have been used for [17] constructing software defect prediction models [17].

From these techniques, supervised learning[17] technique is widely used to be highly appropriate for anticipating software faults if ample quantity of previous defect data is accessible[17]. Machine Learning technique has loomed as compelling approach to anticipate the presence of defects in software components. Various algorithms based on machine learning technique are: Neural networks, Regression Analysis, Case Based Reasoning, Support Vector Machines, Fuzzy Classification, and Bayesian Belief Networks [17].

Existing studies in software defect prediction models emphasis on anticipating faults from these aspects:

- *Number of Faults [17]*: It anticipates the number of defects which are present in software components.
- *Classification [17]*: These helps to anticipate defects which are in a software [17] component. They divide modules into faulty [17] modules as well as non-faulty modules [17].

A. Neural Networks role in Software Defect Prediction[17]

Defect prediction models[17] are premised upon interrelationship between software metrics and fault-proneness[17]. Neural Networks have loomed over the decades as it is a very refined modelling approach that imitate very

convoluted functions as well as non-linear relationships [17] which are impossible to imitate using further approaches. They are [17] more useful over analytical models as they require only fault history as input [17]. They consists of interconnected nodes known as neurons. These neurons accept signals, process these signals and then finally produce an output signal [17].

The paper[18] describes the case study[18] of various neural-network modelling systems which are developed for the EMERALD system[18]. Neural-network based models are used to determine fault-prone modules[18] for early in development phase so that the risk of various operational problems related to these modules reduces.

V. CONCLUSION

After analyzing various studies related to data mining approaches for software defect prediction, we found that data mining has loomed as one of the approach for predicting software defects. Different models and approaches are studied those have various pros and cons associated with them. The main motive behind this study was to explore various data mining [17] techniques for software defect prediction models [17].

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